Cortical Columns

Contents

1. Cortical columns versus mini-columns

There is a difference between **cortical columns** and **mini-columns**, although they are related concepts in neocortical neuroscience.

1. **Mini-columns (Micro-columns)**:

    - These are the fundamental repeating units of the neocortex, each containing **80–100 neurons**.
    - Mini-columns are about **30–50 microns** in diameter and are organized vertically through the six layers of the neocortex.
    - Each mini-column contains pyramidal neurons, interneurons, and other supporting cells, which are highly interconnected.
    - They are believed to be the basic unit of information processing in the neocortex.

2. **Cortical Columns (Macro-columns or Functional Columns)**:

    - These are larger structures, **composed of multiple mini-columns**.
    - Cortical columns can be **several hundred microns** in diameter (typically **300–500 microns**).
    - They are thought to represent functionally distinct units, where neurons within a column process similar types of information (e.g., orientation columns in the visual cortex that respond to specific edge orientations).
    - They help organize sensory and motor processing in a modular fashion.

Relationship:

- Mini-columns are the **smallest building blocks** of cortical organization.
- Cortical columns **group multiple mini-columns together** into a larger functional unit.

In summary, **mini-columns are the microscopic units**, while **cortical columns are the macroscopic, functional aggregations of mini-columns** that work together to process information.

The relationship between **mini-columns** and **cortical columns** is hierarchical: **mini-columns** are the **smallest structural units**, while **cortical columns** are larger **functional units** composed of multiple mini-columns.**Mini-Columns and Cortical Columns in the Visual Cortex**

Mini-Columns: Fine-Grained Processing Units

- Mini-columns (sometimes called **micro-columns**) contain around **80–100 neurons** and are **very narrow (30–50 microns in diameter)**.
- They are **organized vertically** through the six layers of the neocortex.

- In the **visual cortex (V1)**, each mini-column is believed to **respond to a very specific aspect of visual information**, such as a small part of the visual field.
- A **single mini-column** is generally selective to an edge **orientation** at a specific **point in the visual field**.
    - For example, one mini-column might respond to a **vertical line** in a small part of the visual field, while another mini-column nearby might respond to a **slightly tilted line**.

Cortical Columns: Larger Functional Units

- A **cortical column** (sometimes called **macro-column**) is **larger (300–500 microns in diameter)** and is **made up of multiple mini-columns**.
- In the **visual cortex**, cortical columns group mini-columns that process similar features together.
- **Example: Orientation Columns**
    - Within a **cortical column**, there are many mini-columns that each respond to a **slightly different orientation** of an edge.
    - A **single cortical column** represents a **complete set of orientations** for a given small **region of the visual field**.
- **Example: Ocular Dominance Columns**
    - In V1, cortical columns are also arranged into **ocular dominance columns**, where some columns preferentially respond to input from the **left eye** and others to the **right eye**.
    - Each **ocular dominance column** contains many **mini-columns** that analyze different **features (e.g., orientation, spatial frequency, color)** from one eye.

Summary of the Relationship

Feature	Mini-Column	Cortical Column
Size	~30–50 microns	~300–500 microns
Neurons	~80–100	Several hundred to thousands
Function	Encodes fine details (e.g., one specific edge orientation at a specific location)	Groups mini-columns into functional units (e.g., full set of orientations for a location, ocular dominance)
Example in V1	A single mini-column responds to a **specific edge orientation** (e.g., vertical) at a **specific retinal location**	A full cortical column encodes **all orientations** for that **same location** and might belong to an **ocular dominance column**

Thus, mini-columns act as **fine-grained feature detectors**, while cortical columns serve as **larger organizational units that integrate information from multiple mini-columns** to process visual stimuli more comprehensively.

2. Activity on mini-columns

Not all neurons within a mini-column are active at the same time when the mini-column is engaged. Instead, activation within a mini-column is **dynamic, sparse, and context-dependent**. Here's why:

Layer-Specific Activity

- A mini-column spans **all six layers of the neocortex**, but different layers have **different roles** and are **activated at different times** depending on the type of processing.

INPUT

  - **Layer 4 (input layer)**: Typically receives sensory input (e.g., from the lateral geniculate nucleus in the visual cortex).

PROCESS

  - **Layer 2/3 (intracortical processing layer)**: Neurons process and integrate information within the column and with neighboring columns.

OUTPUT

  - **Layer 5/6 (output layers)**: These layers send signals to subcortical structures or other cortical areas.

- If a mini-column is **responding to a stimulus**, neurons in **Layer 4** might be highly active, while neurons in **Layer 5** may be preparing an output but not firing yet.

Sparse Activation

- **Only a subset of neurons in the mini-column fire at any given moment**.
- The brain operates in a **sparse coding** regime, meaning that only a small percentage of neurons in any given area are active simultaneously.
- This sparsity **increases efficiency** (reduces energy consumption) and **enhances signal-to-noise ratio**.

Inhibitory Control

- Mini-columns contain **excitatory pyramidal neurons** but are also surrounded by **inhibitory interneurons**.
- These interneurons **regulate which neurons fire** and prevent **excessive activity**, ensuring that only the most relevant neurons in the mini-column remain active.
- This means that even if a mini-column is responding to a stimulus, **not all excitatory neurons will fire due to local inhibitory control**.

Task and Context Dependence

- The exact pattern of firing within a mini-column depends on **what the brain is doing**.

- For example, in the **visual cortex**, a mini-column responding to a vertical edge in normal light might activate **a different set of neurons** compared to the same mini-column processing a **blurred or moving edge**.

Thalamocortical and Corticocortical Modulation

- Mini-columns receive input from **thalamic** and **other cortical areas**, which can **amplify or suppress** their activity dynamically.
- Depending on **attention, expectation, and learning**, some neurons within a mini-column may be **more excitable** than others.

Conclusion

- A mini-column functions as a **coherent processing unit**, but **not all of its neurons fire simultaneously**.
- Instead, activation within a mini-column is **layer-specific, sparse, and dynamically modulated** by both internal and external influences.

This **sparse and selective activation** allows the brain to **efficiently process information** while **avoiding unnecessary energy expenditure and noise**.

3. Interneurons and inhibitory activity

How Interneurons Inhibit Activity in Mini-Columns

Interneurons in the neocortex play a crucial role in **regulating neural activity** through **inhibition**, ensuring that information processing remains efficient and controlled. They do this by **modulating the firing of excitatory pyramidal neurons** through **inhibitory neurotransmitters**, primarily **GABA (gamma-aminobutyric acid)**.

Interneurons are activated by incoming signals from:

- **Excitatory pyramidal neurons (local or long-range)** that provide direct or indirect input.
- **Thalamic input** (especially in sensory areas like the visual cortex).
- **Other inhibitory interneurons**, which can form complex inhibitory networks.

Types of Inhibition in Mini-Columns

There are three main types of inhibition that interneurons mediate in mini-columns:

1. **Feedforward Inhibition** (Fast Control of Excitation)
    - When a sensory input excites a pyramidal neuron in **Layer 4**, it simultaneously excites an interneuron.
    - The interneuron **quickly inhibits nearby pyramidal neurons**, ensuring that only the most strongly activated ones respond.
    - This helps **sharpen responses** and **prevent runaway excitation**.
    - Example: In the **visual cortex (V1)**, when a mini-column responds to a vertical edge, feedforward inhibition ensures that nearby mini-columns **respond less strongly**, enhancing contrast.
2. **Feedback Inhibition** (Regulating Network Activity)
    - Higher-layer pyramidal neurons in **Layer 2/3 or Layer 5** send signals **back** to interneurons.
    - The interneurons then inhibit the pyramidal neurons within the mini-column or surrounding area.
    - This creates **self-regulation**, preventing overactivation.
    - Example: In decision-making circuits, feedback inhibition helps prevent **prolonged firing** when attention shifts to a new stimulus.
3. **Lateral Inhibition** (Contrast Enhancement & Competition)
    - Excited pyramidal neurons in one mini-column **activate inhibitory interneurons** that **suppress activity in adjacent mini-columns**.
    - This helps the brain **enhance contrast** by making the **strongest signals more dominant** while suppressing weaker or irrelevant ones.

- Example: In the **visual cortex**, when one mini-column detects a **high-contrast edge**, it **inhibits neighboring columns** responding to weaker or less relevant stimuli.

Types of Interneurons in Mini-Columns

There are different classes of **GABAergic inhibitory interneurons**, each playing a unique role in controlling pyramidal neurons.

1. Parvalbumin (PV) Interneurons – Fast, Strong Inhibition

- **Example: Basket Cells, Chandelier Cells**
- **Target**: Cell bodies and axon initial segments of pyramidal neurons.
- **Function**: Provide **strong, fast-spiking inhibition**, limiting when pyramidal neurons can fire.
- **Activation**: Excited by pyramidal neurons via AMPA synapses, then rapidly shut down nearby excitatory activity.
- **Role in Mini-Columns**:
    - Ensure that only the most **salient** stimuli are processed.
    - Control **precise timing of neuronal firing** (important for synchronization).

2. Somatostatin (SOM) Interneurons – Dendritic Modulation

- **Example: Martinotti Cells**
- **Target**: Dendrites of pyramidal neurons.
- **Function**: Regulate **how much input a pyramidal neuron receives** from other neurons.
- **Activation**: Activated by pyramidal neurons in deeper layers, they provide feedback inhibition.
- **Role in Mini-Columns**:
    - Control how **strongly** a neuron integrates incoming signals.
    - Important for **tuning** responses and preventing excessive firing.

3. VIP (Vasoactive Intestinal Peptide) Interneurons – Disinhibition

- **Target**: Other interneurons (PV and SOM cells).
- **Function**: Suppress inhibition, allowing pyramidal neurons to fire **more easily** when needed.
- **Activation**: Can be triggered by top-down signals (e.g., attention, learning).
- **Role in Mini-Columns**:
    - **Enable flexible learning** by temporarily reducing inhibition when necessary.
    - **Amplify signals during attention**, helping specific mini-columns become more active.

How Interneurons are activated

Interneurons themselves are activated by a combination of **excitatory input from pyramidal neurons** and **thalamic input**. Their activation depends on:

Sensory Inputs (Thalamocortical Projections)

- Sensory signals excite both **excitatory pyramidal neurons** and **inhibitory interneurons** in Layer 4.
- Interneurons quickly suppress weakly responding pyramidal neurons.

Cortico-Cortical Feedback

- Higher cortical areas send **feedback signals** that can either increase or decrease inhibition.
- Example: In **attention**, VIP interneurons inhibit SOM cells, **disinhibiting** specific mini-columns and making them **more responsive**.

Intrinsic Network Activity

- Some interneurons are engaged **spontaneously** to maintain balance, preventing runaway excitation.

Conclusion

- **Interneurons are key regulators of activity within mini-columns**, ensuring that pyramidal neurons do not all fire simultaneously.
- They act through **feedforward, feedback, and lateral inhibition**, creating a **balance between excitation and inhibition**.
- Different types of interneurons (PV, SOM, VIP) **modulate activity at different levels**, ensuring precise control of information processing.
- Their **activation is dynamic**, influenced by sensory input, local pyramidal neurons, and top-down modulation.

This **inhibitory control** is essential for functions like **contrast enhancement in vision, attention selection, and preventing epileptic-like overactivation** in the brain.

Distribution of Inhibitory Neurons across Cortical Layers

Layer 4 (Granular Layer)

- Often called the **primary input layer**, especially in sensory cortices (like V1).
- Rich in **Parvalbumin-positive (PV+) fast-spiking interneurons**, such as **basket cells**, which provide **feedforward inhibition**.
- These interneurons help **shape the initial response to thalamic input** by quickly suppressing nearby pyramidal neurons.

Layers 2/3 (Supragranular Layers)

- Contain **interneurons that mediate feedback and lateral inhibition**.
- Includes:
  - **PV+ basket cells** that inhibit soma of pyramidal neurons.
  - **Somatostatin-positive (SOM+) Martinotti cells**, which target distal dendrites of pyramidal neurons — these are important for **controlling dendritic input integration**.
  - **Vasoactive Intestinal Peptide (VIP+) interneurons**, which **inhibit other inhibitory neurons** (i.e., disinhibition).
- Involved in **horizontal interactions between columns** (lateral inhibition), and **top-down modulation** (e.g., attention, context, learning).

Layer 5 (Infragranular Layer)

- Has many large pyramidal neurons (e.g., **Betz cells**) that project subcortically.
- Also contains:
  - PV+ basket cells providing **strong somatic inhibition**.
  - SOM+ cells modulating **dendritic computation**.
  - VIP+ interneurons involved in **cortical state changes and disinhibition**.
- Layer 5 interneurons help shape the **output** of the cortex and regulate long-range **descending signals**.

Layer 6

- Contains inhibitory interneurons that influence both **local circuits and thalamic feedback loops**.
- Helps regulate **gain control**, **feedback inhibition**, and **top-down modulation** to earlier sensory areas.

Key Point

Every layer has a **specific composition** of **inhibitory neuron subtypes**, and they are **functionally integrated**:

- **Layer 4**: Dominated by **feedforward inhibition** (fast-spiking PV+ cells responding to thalamic input).
- **Layers 2/3 and 5**: Rich in **feedback and lateral inhibition**, and modulation via **VIP→SOM→Pyramidal disinhibition pathways**.
- **Layer 5**: Critical for **output control**, often involved in **recurrent and long-range inhibition**.

Layer	Inhibitory Neurons Present?	Main Functions
2/3	✅ Yes	Lateral & feedback inhibition, disinhibition

Layer	Inhibitory Neurons Present?	Main Functions
4	✅ Yes	Feedforward inhibition from sensory input
5	✅ Yes	Output gating, feedback inhibition
6	✅ Yes	Gain control, corticothalamic modulation

So, to answer directly:
➡️ **Inhibitory neurons are not restricted to Layer 4**. They are **present and functionally crucial in all cortical layers**.

Layers 2/3 Interneurons: Overview

how interneurons in layers 2/3 (the supragranular layers) modulate cortical activity through **inhibition**, especially **feedback inhibition**. You're absolutely right that these interneurons often **inhibit different neuron populations** than those initially activated in **Layer 4**, and you're also right to seek more clarity on what **feedback inhibition** means in this layered cortical context.

Recap of Layer Functionality

- **Layer 4** is typically the **main input layer**, especially in primary sensory cortex (e.g. V1), receiving **feedforward sensory input** from the **thalamus**.
- Neurons in Layer 4 then **project vertically to neurons in Layers 2/3**, which are heavily involved in **integration, horizontal communication, and transmitting information to other cortical areas**.

Layers 2/3

In **Layers 2/3**, there are three main classes of inhibitory interneurons:

Interneuron Type	Marker	Main Target	Function
PV+ (Parvalbumin)	Basket Cells	Soma/axon initial segment of pyramidal neurons	Precise spike timing, inhibition of firing
SOM+ (Somatostatin)	Martinotti Cells	Apical dendrites of pyramidal neurons	Dendritic inhibition, integration control
VIP+ (Vasoactive Intestinal Peptide)	VIP Interneurons	Other interneurons (esp. SOM+)	Disinhibition (turning off inhibition)

Feedback Inhibition in Layers 2/3: Explained

What is Feedback Inhibition?

- **Feedback inhibition** occurs when **excitatory pyramidal neurons activate inhibitory interneurons**, which in turn **inhibit the same or nearby pyramidal neurons**.

- It's a **self-regulatory loop** that limits the duration or strength of cortical excitation.

Circuit Example (Visual Cortex):

Let's say a vertical edge in the visual field activates a **mini-column**:

1. **Layer 4** receives thalamic input → activates local pyramidal neurons.
2. These neurons project **up to Layer 2/3**, exciting pyramidal neurons there.
3. The Layer 2/3 pyramidal neurons **send collateral excitatory signals to local interneurons**:
    - **PV+ basket cells** → inhibit the **somata** of nearby pyramidal neurons.
    - **SOM+ Martinotti cells** → inhibit **apical dendrites**, shaping integration of further inputs.
4. These interneurons now **suppress or delay** further firing of pyramidal neurons in **Layer 2/3**, especially those that were just activated.
    - This limits the **spread or duration of excitation**, which is **feedback inhibition**.

So yes, **Layer 2/3 interneurons inhibit pyramidal neurons in Layer 2/3, not Layer 4.**
The target pyramidal neurons might be the **same ones** that activated the interneurons (self-inhibition), or **neighboring ones** (local inhibition).

Feedforward x Feedback Inhibition

Feature	Feedforward Inhibition	Feedback Inhibition
Source of Excitation	**Thalamus** or external input	**Local pyramidal neurons**
Activated Interneurons	Often in **Layer 4**	Often in **Layer 2/3**
Target	Layer 4 and 2/3 pyramidal cells	Layer 2/3 pyramidal cells
Function	Sharpen response to external input	Limit or fine-tune ongoing local activity

Functional Role of Feedback Inhibition in 2/3

- **Maintains excitation/inhibition (E/I) balance** — prevents runaway activity.
- **Sharpens spatial and temporal resolution** — only the most relevant mini-columns stay active.
- **Supports attention and learning** by allowing flexible modulation via **disinhibition**.
- **Contributes to oscillations and synchrony** — especially gamma rhythms, which involve PV+ cells.

Bonus: Experimental Findings

- In vivo calcium imaging and optogenetics (e.g. Tremblay et al., 2016; Karnani et al., 2016) show that:

- **PV+ cells fire rapidly after pyramidal neurons**, consistent with feedback inhibition.
- **SOM+ cells** are more involved in **delayed, dendritic modulation**, especially during sustained or top-down activity.
- **VIP+ cells** can be activated by **behavioral state or attention**, and they **suppress SOM+ cells**, thus **disinhibiting pyramidal neurons**.

Summary Answer to Your Question

> Are the pyramidal neurons inhibited by Layer 2/3 interneurons the same ones activated by Layer 4 input?

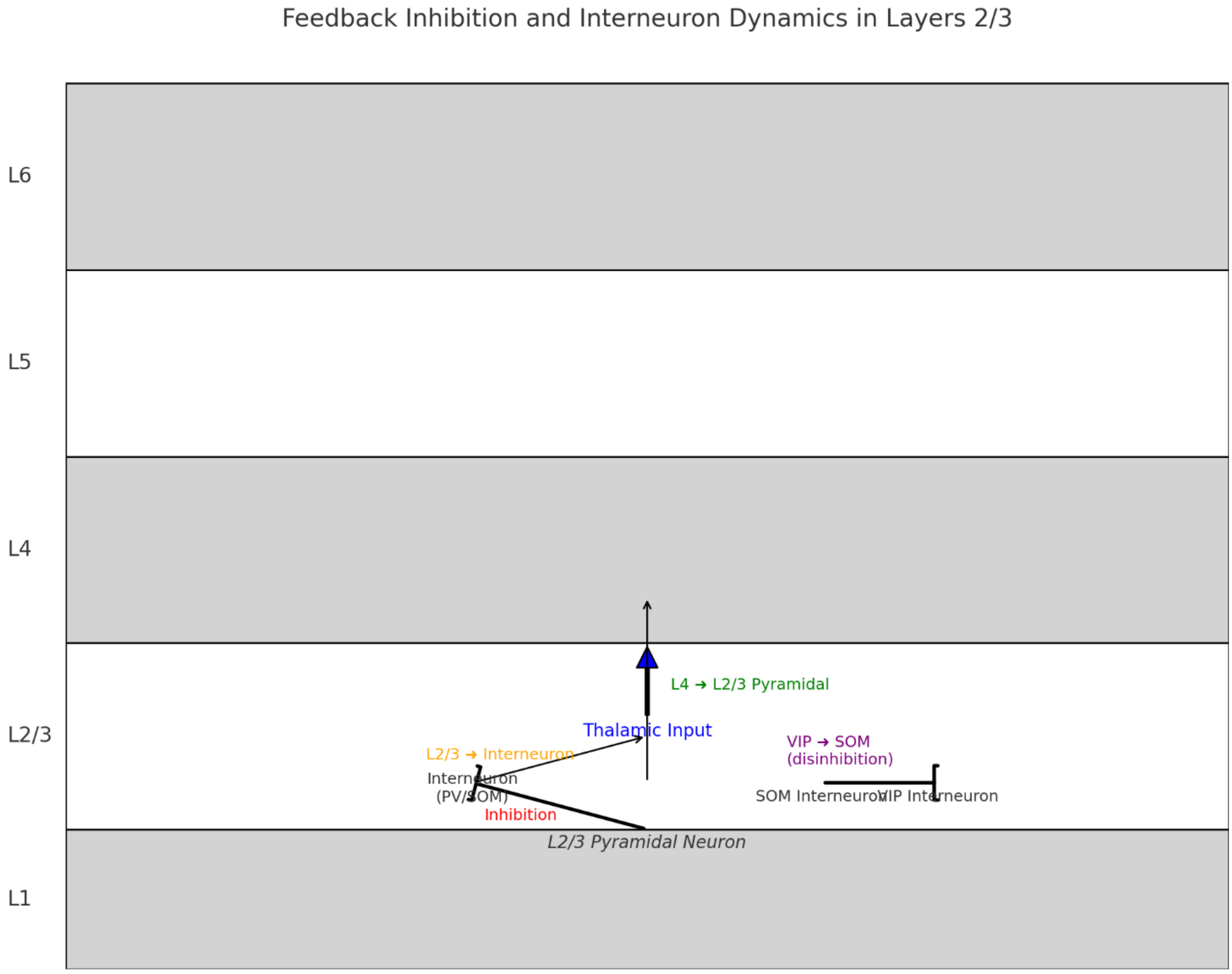
Not directly.

- **Layer 4** activates **Layer 2/3 pyramidal neurons**, and those then **activate local interneurons**.
- **Layer 2/3 interneurons inhibit pyramidal neurons in the same or neighboring columns** in **Layer 2/3**, not Layer 4.
- This constitutes **feedback inhibition**, because it's **driven by the activity of excitatory neurons**, rather than by the incoming input itself.

This process **regulates the local network**, enforces **competition among mini-columns**, and ensures **selective processing** of sensory information.

Would you like a diagram to help visualize this?

Here's a visual diagram illustrating how **feedback inhibition** and **interneuron dynamics** operate in **Layers 2/3** of the neocortex:



Key Pathways in the Diagram:

- **Blue arrow**: Sensory input from the **thalamus** targets **Layer 4**.
- **Green arrow**: Layer 4 **excites pyramidal neurons in Layer 2/3**.
- **Orange arrow**: These **pyramidal neurons activate local inhibitory interneurons** (PV or SOM).
- **Red line**: Interneurons **inhibit the same or nearby pyramidal neurons** — this is **feedback inhibition**.
- **Purple inhibition**: **VIP interneurons inhibit SOM cells**, leading to **disinhibition** (pyramidal neurons become more excitable).

Let me know if you'd like the same logic applied to other cortical layers or a labeled version for use in presentations or reports!

Lateral Inhibition in Layers 2/3

Lateral inhibition is when neurons in one mini-column (or cortical column) **inhibit neurons in neighboring columns**, creating a **"winner-takes-most"** dynamic:

- The **most strongly activated mini-columns suppress the activity of surrounding ones**.

- This allows the cortex to focus computational resources on the **most relevant or salient stimuli**.
- It's essential for things like **edge detection, orientation selectivity, attention, and working memory**.

Lateral inhibition in Layers 2/3 involves horizontal projections that extend beyond a single column.

- Pyramidal neurons in Layers 2/3 **send long-range horizontal axons** that can reach **up to several millimeters** across the cortex.
- These projections often form **patchy or stripe-like connection patterns**, seen clearly in **visual cortex (V1)** as **orientation-specific networks**.
- Along the way, these axons **activate local inhibitory interneurons** in **neighboring columns**, particularly **PV+ or SOM+ interneurons**.
- This leads to **inhibition of pyramidal neurons in those neighboring columns** — this is the core of **lateral inhibition**.

So, while the **interneuron itself is local**, the **excitation that activates it comes from pyramidal neurons in distant columns**.

Which Interneurons Are Involved?

1. PV+ (Parvalbumin-positive) Basket Cells

- **Receive horizontal input** from pyramidal neurons of other columns.
- Inhibit **somata** of pyramidal neurons in their own local mini-column.
- Important for **fast, precise lateral inhibition** — sharpening timing and selectivity.

2. SOM+ (Somatostatin-positive) Martinotti Cells

- Also receive lateral excitatory input from distant pyramidal cells.
- Inhibit **apical dendrites** of pyramidal neurons.
- Useful for **gating dendritic integration** and suppressing **longer-lasting inputs** (e.g., top-down feedback).

3. Less Involved: VIP+ Interneurons

- Typically mediate **disinhibition**, not lateral inhibition.
- They modulate the local inhibitory balance, but are **less directly involved in lateral inhibition** across columns.

Target Neurons of Lateral Inhibition

Lateral inhibition affects mainly:

- **Pyramidal neurons in neighboring columns**:

    - Inhibited at the **soma (PV+)** or **dendrites (SOM+)**.
- The goal is to **suppress pyramidal output** in columns that are **less strongly activated** by the stimulus.

This supports **columnar competition** and helps the cortex make **sharp decisions** about what to represent.

Example: Visual Cortex (V1)

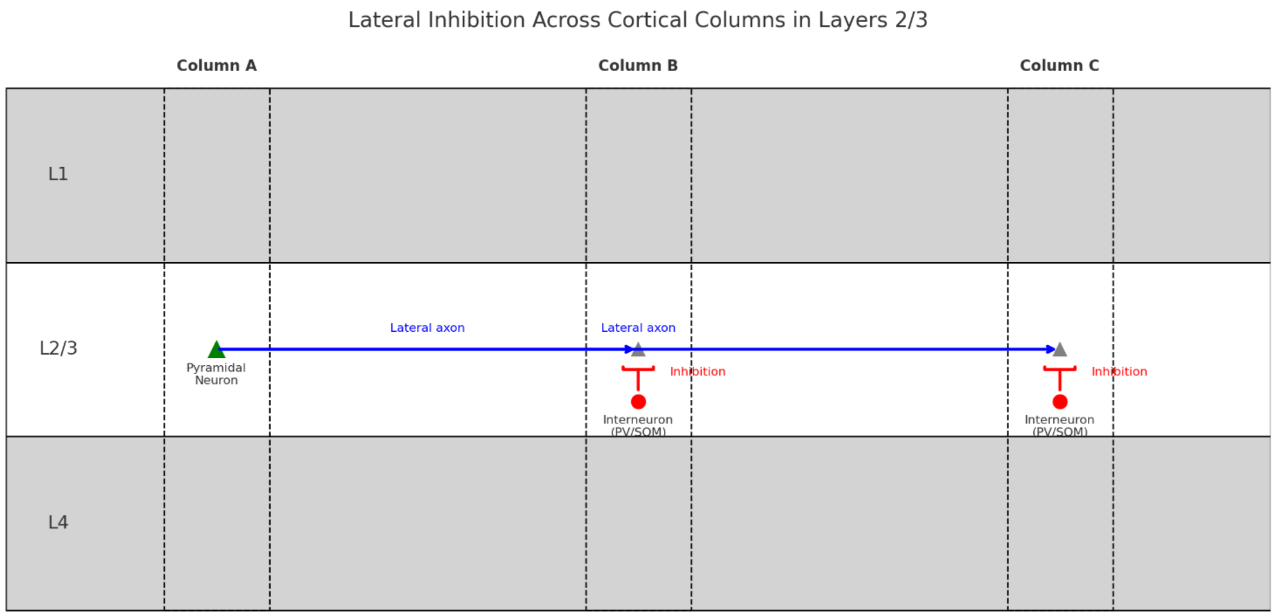
- Suppose a vertical edge activates a group of mini-columns tuned to that orientation.
- These columns **laterally inhibit** columns nearby that are tuned to **slightly different orientations** (e.g., 15° off).
- This **sharpening of tuning curves** is due to:
    - **Horizontal projections** from active pyramidal neurons,
    - Activating **interneurons** in other mini-columns,
    - Leading to **inhibition of their pyramidal cells**.

➡️ As a result, the **strongest representation is emphasized**, and weaker or ambiguous responses are **suppressed**.

Key Takeaways

Feature	Description
Lateral inhibitory connections	Extend across columns, can reach millimeters across cortex
Origin of inhibition	Local PV+ and SOM+ interneurons activated by long-range pyramidal axons
Target	Pyramidal neurons (mostly in Layers 2/3), both soma and dendrites
Function	Sharpen tuning, enhance contrast, enforce competition, focus cortical resources
Present in	V1 (orientation), S1 (touch), A1 (frequency), and association areas (working memory, attention)

Here is the diagram showing **lateral inhibition across cortical columns** in **Layers 2/3**:



How to Read the Diagram:

- **Column A**:
  - Contains a **pyramidal neuron** (green triangle) in Layer 2/3.
  - It sends **horizontal axons** (blue arrows) to distant columns.
- **Columns B and C**:
  - These axons **excite local inhibitory interneurons** (red circles, PV or SOM).
  - These interneurons **inhibit local pyramidal neurons** (gray triangles) via **red inhibitory signals**.

Functional Meaning:

- When Column A is strongly activated (e.g. by a preferred stimulus), it **suppresses neighboring columns** through **lateral inhibition**.
- This improves **contrast, selectivity, and competition** across the cortex.

4. Top-down connections

Top-down connections are projections from **higher-order cortical areas to lower-order areas**, and they play a key role in:

- **Expectation**
- **Attention**
- **Working memory**
- **Perceptual inference**

These connections let the brain **predict, shape, or reinterpret** sensory input based on **context, goals, or past experience**.

Where Do Top-Down Projections Come From?

Top-down projections **originate mostly from Layers 2/3 and 5** of **higher-order cortical areas** (like prefrontal cortex, association areas) and target:

- **Layers 1 and 6** of lower-order areas.
- Sometimes also **Layer 2/3** of lower areas.

Typical Path:

- A **higher cortical area** (e.g. prefrontal cortex, PFC) sends **top-down signals** to a **sensory area** (e.g. visual cortex, V1).
- These axons **terminate in superficial layers**, especially **Layer 1**, where they **modulate the apical dendrites** of pyramidal neurons in **Layer 2/3 and 5**.
- **This is how internal models influence the interpretation of sensory input.**

Is Top-Down the Same as L3 ➜ L5 Connections?

Not exactly.

- Connections **within a cortical column** from **Layer 3 to Layer 5** are usually **local, vertical, bottom-up** (input integration ➜ output preparation).
- **Top-down** refers to **long-range projections** across **different cortical regions**, not local connections.

That said, **Layer 3 and Layer 5** pyramidal neurons in **higher areas** both contribute axons to top-down pathways, but they target **earlier areas** (e.g., L1 of V1).

How Do Top-Down Connections Support Inference?

Now we get to the exciting part — **predictive coding** and **inference**.

Predictive Coding (Simplified Explanation)

1. **Bottom-up input** (from thalamus or earlier cortex) arrives at **Layer 4** of a sensory area.

2. **Top-down input** (from higher cortex) arrives at **Layer 1**, influencing the **apical dendrites** of **Layer 2/3 and 5 pyramidal neurons**.
3. These two inputs are **compared**:
   - If the **bottom-up input matches the top-down prediction**, pyramidal neurons are **modulated but not strongly activated**.
   - If there's a **mismatch**, a **prediction error** is generated, and pyramidal neurons fire to **update the higher areas**.
4. This **error signal** propagates **up the hierarchy**, helping the brain learn and refine its models.

Why Apical Dendrites Matter

- The **apical dendrites** of pyramidal neurons in L2/3 and L5 reach into **Layer 1**, where top-down signals arrive.
- These top-down signals can **modulate the neuron's likelihood of firing**, depending on whether they match the bottom-up input at the soma.

This allows **contextual modulation**:

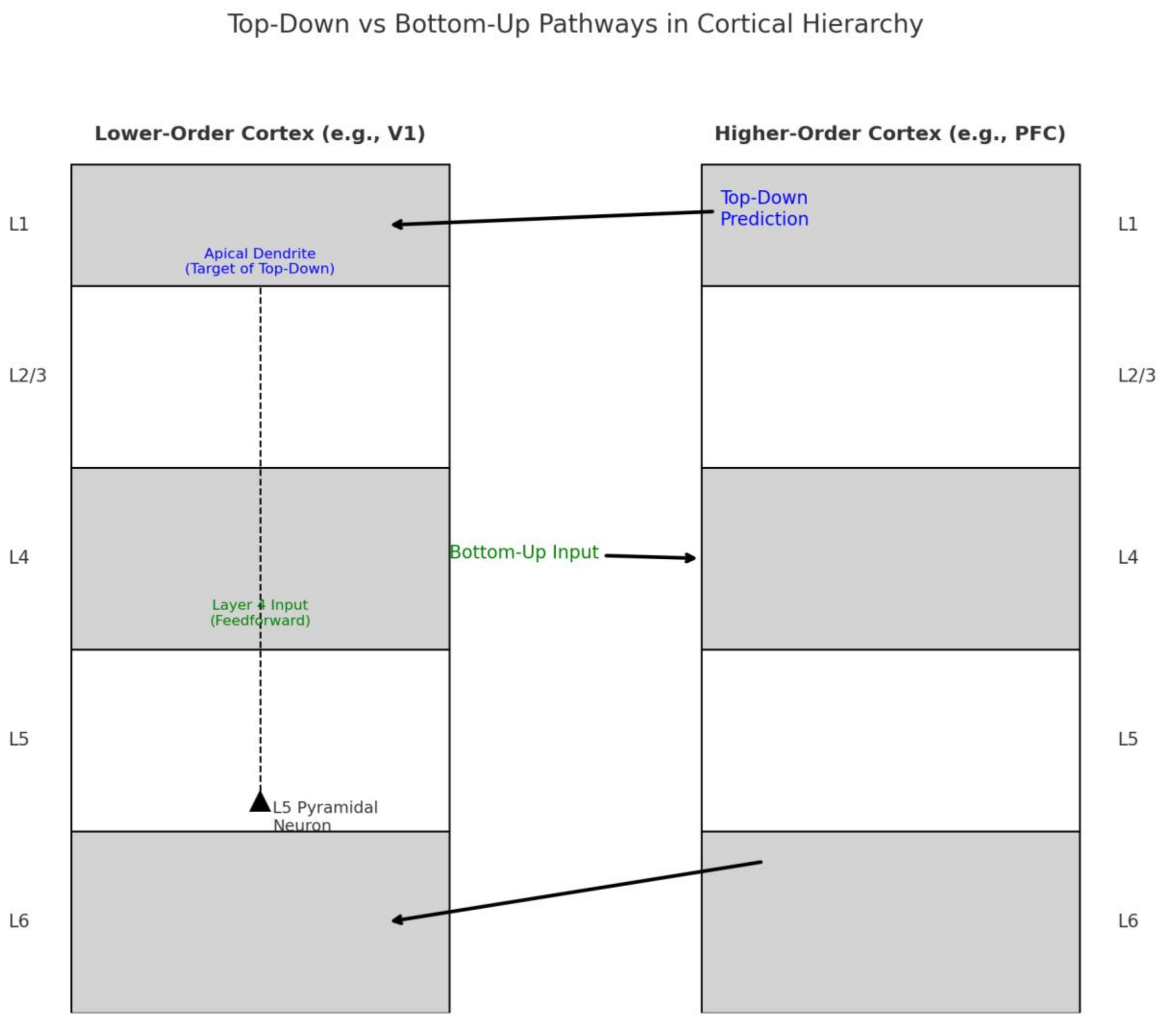
- A weak visual input might be **boosted** if it matches a strong expectation.
- A strong but irrelevant input might be **ignored** if it conflicts with current goals.

Summary of the Mechanism

Feature	Top-Down Pathway
Origin	Layers 2/3 or 5 of **higher cortical areas**
Target	Layer 1 (apical dendrites) and Layer 6 of **lower areas**
Function	Modulate, predict, interpret bottom-up input
Related Concept	**Predictive coding**, perceptual inference
Local L3 ➔ L5	Vertical integration within a column, **not top-down** per se

Key Insight

> Top-down connections enable the brain to **infer the causes of sensory inputs**, not just react to them.
> This allows **perception to be shaped by experience, attention, memory, and expectation**.



Key Elements:

- **Left side**: Lower-order cortex (e.g. primary visual cortex, V1)
- **Right side**: Higher-order cortex (e.g. prefrontal cortex, PFC)

Bottom-Up Pathway:

- Sensory input arrives in **Layer 4** of the lower cortex.
- Activates **Layer 2/3 pyramidal neurons**, which send signals **up** to the higher cortex.

Top-Down Pathway:

- Comes from **Layers 2/3 or 5** of the higher cortex.
- Projects to:
    - **Layer 1** of the lower cortex (modulating **apical dendrites** of pyramidal neurons).
    - **Layer 6**, potentially influencing deep feedback loops.

Functional Role:

- **Bottom-up**: Delivers raw sensory input.

- **Top-down**: Delivers **context, expectation, memory, or goals**, which modulate how input is interpreted — enabling **inference** and **predictive coding**.

Let me know if you'd like to expand this with prediction error signals or the role of interneurons in these dynamics!

5. Predictive Coding

Predictive coding is a **theory of brain function** that says:

> The brain is constantly generating predictions about incoming sensory input and only processes the **difference between prediction and reality** — this is the **prediction error**.

It's like the cortex is trying to "guess" what the world is doing and only corrects itself when it's wrong.

Cortical Implementation (Simplified Hierarchy)

Cortical Level	Signal
Higher-order cortex	Sends **predictions** downward (top-down)
Lower-order cortex	Sends **prediction errors** upward (bottom-up)

- **Top-down projections** deliver **expectations** about the world.
- **Bottom-up inputs** carry **what actually happened**.
- **Mismatch = error**, which is sent **back up** the hierarchy to adjust the model.

Where Do Predictions and Errors Flow?

Top-Down Predictions:

- Originate in **Layers 5 and 2/3** of higher-order cortex.
- Target:
  - **Layer 1** (apical dendrites of pyramidal neurons)
  - **Layer 6** (modulatory deep connections)

Bottom-Up Errors:

- Generated in **Layer 2/3 and Layer 5** of lower cortex.
- Sent upward via **ascending pathways** to update models in higher areas.

What Exactly Is a Prediction Error Signal?

A **prediction error** is a signal that represents the **difference between expected input and actual input**.

In cortical terms:

- If a **pyramidal neuron's soma** receives **sensory input** (e.g. from Layer 4),
- And its **apical dendrite** receives a **matching prediction** (e.g. from Layer 1),
- Then the neuron is **less likely to fire** → "I expected this, no error."
- If the inputs **mismatch**, the neuron **fires more**, signaling "surprise" → **prediction error**.

Role of Interneurons in Predictive Coding

Now here's where it gets super interesting.

1. SOM+ Interneurons – *Gate Top-Down Predictions*

- SOM (Somatostatin-positive) interneurons **inhibit apical dendrites** of pyramidal neurons.
- They **control whether top-down signals in Layer 1 reach the soma**.
- Can **suppress irrelevant predictions**, allowing only the **strongest or most relevant expectations** to influence perception.

They regulate how much top-down context reaches perception.

2. PV+ Interneurons – *Gate Bottom-Up Signals & Precision*

- PV (Parvalbumin-positive) interneurons **inhibit the soma** of pyramidal neurons.
- Control the **gain** (sensitivity) of bottom-up inputs.
- If a prediction is **confident**, PV interneurons **suppress weak mismatches** — lowering sensitivity.
- If uncertainty is high, **less PV inhibition** lets more error signals through.

They regulate **how precisely** the brain listens to sensory input.

3. VIP+ Interneurons – *Enable Disinhibition*

- VIP (Vasoactive Intestinal Peptide) interneurons **inhibit SOM+ cells**, leading to **disinhibition** of apical dendrites.
- Under attention, learning, or surprise, VIP neurons **enhance the influence of top-down predictions**.

They allow **flexible routing** of prediction signals based on cognitive state.

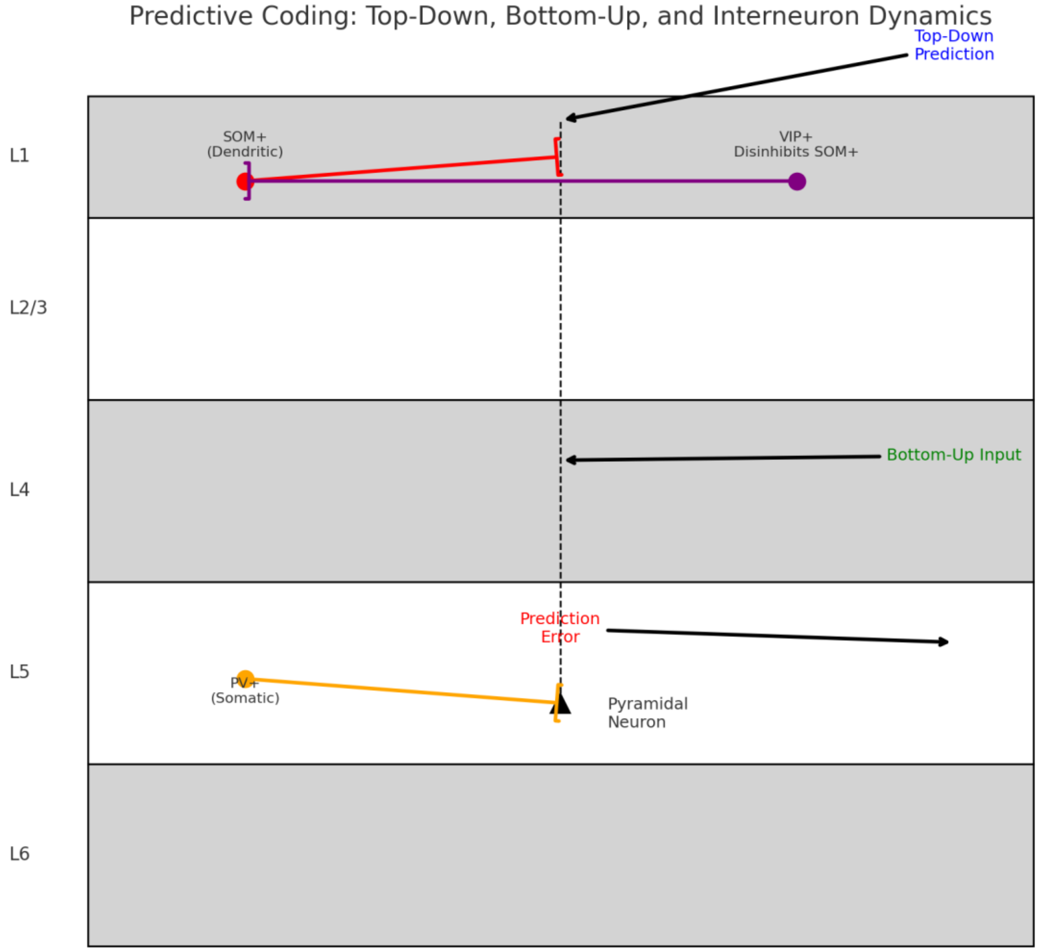
Visual Summary

Interneuron	Target	Role in Predictive Coding
SOM+	Apical dendrites	Gate top-down input (context, expectation)
PV+	Soma	Regulate bottom-up signal gain (error precision)
VIP+	Inhibit SOM+	Disinhibit pyramidal dendrites (modulate top-down impact)

Why This Matters

- Predictive coding is not just a theory — it's **implemented through real neural microcircuits**.
- The **balance of inhibition and disinhibition**, largely controlled by interneurons, **determines how we perceive and learn**.

- Errors are not failures — they are the **fuel of learning**.
- This mechanism allows the brain to **stay efficient**, focusing on **unexpected or important information**.



What It Shows:

Top-Down Prediction

- Arrives at **Layer 1**, targeting the **apical dendrites** of the pyramidal neuron.
- May be **gated by SOM+ interneurons** (red), which can inhibit these dendrites.

Bottom-Up Input

- Enters via **Layer 4** and activates the **soma** of pyramidal neurons.
- **PV+ interneurons** (orange) inhibit the soma, controlling the **precision/gain** of sensory input.

Prediction Error

- If top-down and bottom-up signals don't match, a **prediction error** is generated (red arrow) and sent **up the hierarchy**.

VIP+ Interneurons

- Inhibit SOM+ interneurons, creating **disinhibition** of the dendrites.
- This allows **top-down signals** to exert stronger influence during **attention or learning**.

This circuit illustrates how the brain **balances expectations with reality**, and how **interneurons dynamically regulate** which signals dominate perception and learning.

6. Learning

> **Is Hebbian learning still relevant? And how does it fit with predictive coding and cortical processing?**

Yes, Hebbian learning is still valid — but it's now understood as **one component** of a much **richer, layered model** of learning in the brain.
It fits **complementarily** with **predictive coding**, especially when we consider **plasticity rules, synaptic modulation,** and **interneuron control**.

What Is Hebbian Learning?

The classic principle:

> **"Neurons that fire together wire together."**
> (Original formulation by Donald Hebb, 1949)

This means that if **presynaptic neuron A** consistently contributes to the activation of **postsynaptic neuron B**, the **synaptic strength between them increases**.

This leads to:

- Strengthening of correlated pathways.
- Formation of **cell assemblies** (groups of co-active neurons).
- Foundations for **associative memory**, **pattern completion**, and **learning sequences**.

Hebbian learning is the biological analogue of **correlation-based learning**.

Is Hebbian Learning Still Accepted?

Yes — but with **refinements**. Here's what neuroscience has discovered:

Hebbian learning is:

- **Supported by decades of experimental evidence**, especially **long-term potentiation (LTP)** in the hippocampus and cortex.
- **Fundamental to synaptic plasticity**, particularly during **development**, **memory encoding**, and **skill acquisition**.

But modern understanding extends it:

- **Hebbian learning needs regulation** (to avoid runaway excitation).

- Must be **context-sensitive** — not all co-firing should lead to strengthening.
- **Inhibitory neurons**, **neuromodulators** (like dopamine, acetylcholine), and **timing (spike-timing dependent plasticity, STDP)** all modulate Hebbian rules.

How Hebbian Learning Fits into Cortical Circuits (and Predictive Coding)

Here's how it all connects:

1. Prediction Errors Drive Hebbian Plasticity

- In predictive coding, **mismatches between prediction and input** generate **error signals**.
- These errors **drive learning** — the system updates synaptic weights to **reduce future error**.
- Hebbian learning can **strengthen the synapses** that contributed to reducing the error.

So: **Prediction errors tell the brain *when* to learn, and Hebbian learning tells it *what* to learn.**

2. Layer-Specific Hebbian Plasticity

Different layers have different learning functions:

Layer	Activity	Learning Process
L4 → L2/3	Input processing	Sensory associations, feature binding (Hebbian)
L2/3 → L5	Integration → output	Decision/response learning (Hebbian + error modulation)
L1 (top-down)	Context modulation	Model-based adaptation (error-modulated plasticity)

Hebbian learning **increases the strength of co-active connections**, but **interneurons** (PV, SOM, VIP) and neuromodulators **gate** when this plasticity is allowed.

3. Inhibitory Control of Hebbian Plasticity

- PV+ interneurons **limit when and where pyramidal neurons fire**, thus **constraining plasticity**.
- VIP+ interneurons can **disinhibit** learning in selective conditions (e.g. during attention or reward).
- This allows **task-relevant plasticity**, preventing "accidental Hebbian learning" from noise.

In other words, **Hebbian plasticity is permitted — but only in the right context**.

4. Spike-Timing Dependent Plasticity (STDP)

This is a **refinement of Hebb's rule**:

- If a **presynaptic spike precedes a postsynaptic spike** within ~10 ms → **synapse strengthens (LTP)**.
- If the **postsynaptic spike comes first → synapse weakens (LTD)**.

STDP is experimentally observed and aligns **beautifully with predictive coding**:

- **Timing encodes causality** — a key part of model learning.

! ! !

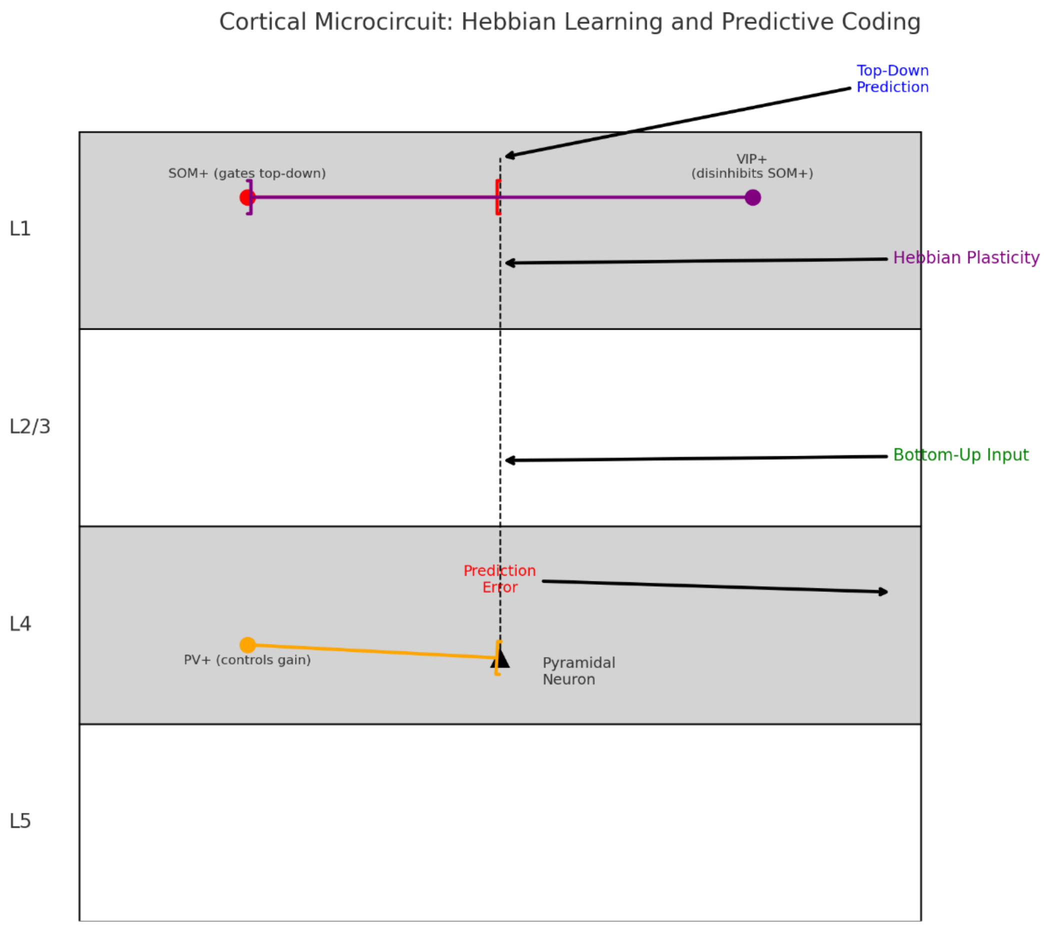
Summary

Concept	Role in Cortical Learning
Hebbian Learning	Core mechanism for strengthening synapses between co-active neurons
Predictive Coding	Provides a framework for how the brain minimizes error through feedback and feedforward signals
Error Signals	Trigger updates to internal models and guide where Hebbian plasticity occurs
Interneurons	Regulate when/where Hebbian learning happens by gating activity and plasticity
Neuromodulators	Set the global state (e.g., attention, novelty, reward) that enables or disables learning

Final Takeaway

Hebbian learning is **still valid and essential**, but it **does not operate in isolation**.
In the cortex, it works in **dynamic interplay** with:

- **Top-down predictions**
- **Error-driven learning**
- **Inhibitory control**
- **Neuromodulatory signals**



Key Components:

- **Bottom-Up Input**: Enters at **Layer 4**, activates the pyramidal neuron at the soma.
- **Hebbian Plasticity**: Strengthens connections between **Layer 4 and Layer 2/3** when pre- and postsynaptic neurons are co-active.
- **Top-Down Prediction**: Arrives in **Layer 1**, modulating the **apical dendrites** of the pyramidal neuron.
- **Prediction Error**: If top-down and bottom-up signals mismatch, an error signal is generated and sent **up the hierarchy**.

Interneuron Roles:

- **SOM+ (Red)**: Inhibits apical dendrites → gates top-down influence.
- **PV+ (Orange)**: Inhibits soma → regulates bottom-up input precision.
- **VIP+ (Purple)**: Disinhibits SOM+ → allows top-down signals to impact learning during attention or novelty.

This diagram captures how learning, context, and inhibition all come together to form a **dynamic, adaptive circuit** — not just passively recording input, but actively **interpreting, modulating, and learning from it**.

7. STDP - Spike-Timing Dependent Plasticity

Spike-Timing Dependent Plasticity (STDP) is a biologically observed learning rule where the **relative timing of spikes** between pre- and post-synaptic neurons determines whether a synapse is strengthened or weakened.

STDP Rule (Classic Version):

Timing (Δt = t_post - t_pre)	Synaptic Change
$\Delta t > 0$ (pre before post)	**LTP** (Long-Term Potentiation) — strengthening
$\Delta t < 0$ (post before pre)	**LTD** (Long-Term Depression) — weakening

So:

- If **input arrives just before the neuron fires**, the brain **reinforces** that connection.
- If **input arrives after** the neuron already fired, it's considered **irrelevant**, and the connection is **weakened**.

STDP allows learning **causal structure** from experience — it's **temporally sensitive Hebbian learning**.

How STDP Maps Onto Top-Down / Bottom-Up Processing

Let's now plug STDP into the **predictive coding circuit**.

Bottom-Up Inputs (Layer 4 → Soma)

- These carry **sensory evidence** and tend to **arrive slightly earlier**, since they are **driven by the external world**.
- If these inputs **precede pyramidal neuron firing**, **STDP reinforces** them.
- This matches the idea that **correct predictions (matching input)** should reinforce the model.

Top-Down Predictions (Layer 1 → Apical Dendrites)

- These are **generated internally** (based on memory, expectation, etc.).
- They tend to **arrive slightly later** or be more **sustained**.
- If the neuron fires **before the top-down input arrives**, **STDP weakens** the synapse.
- If top-down input arrives **just in time to boost firing**, STDP can **strengthen** those dendritic inputs.

STDP Enables Learning of Temporal Relationships

Imagine this flow:

1. A visual input (bottom-up) activates Layer 4 → causes Layer 2/3 pyramidal neuron to fire.
2. Shortly after, a top-down expectation from a higher-order area arrives at Layer 1.

3. STDP recognizes that the top-down signal **didn't cause** the firing → **no strengthening** (or even weakening).
4. If later trials show **top-down prediction consistently helping the neuron fire**, the **top-down synapse is strengthened**.

So over time, the brain **learns which predictions are useful**, and **updates connections accordingly**.

STDP + Predictive Coding = Learning the Model

Mechanism	Bottom-Up	Top-Down
Signal	Actual sensory data	Internal prediction
Arrival Time	Fast (Layer 4 → soma)	Slightly delayed or sustained (Layer 1 → apical dendrites)
STDP Outcome	Pre-before-post = strengthening	Post-before-pre = weakening (unless timing aligns)
Functional Meaning	Learn cause-effect from world	Learn effective predictions from internal models

Implication

STDP supports predictive coding by:

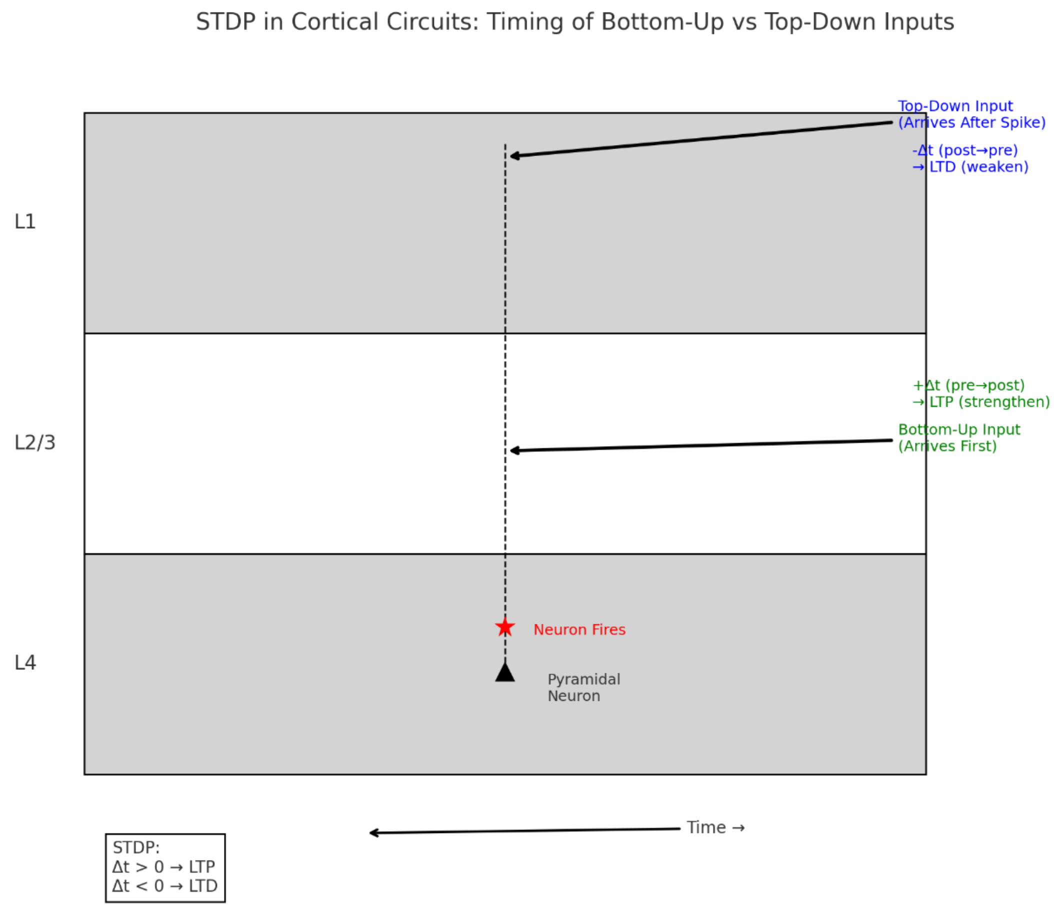
- Reinforcing **bottom-up inputs that truly contribute** to prediction.
- Weakening **irrelevant or mistimed predictions**.
- Updating **both feedforward and feedback pathways** in a biologically plausible way.

In short: **STDP helps the brain *calibrate* its internal model**, one spike at a time.

Bonus: Interneurons Modulate STDP

- **PV+ cells** influence when pyramidal neurons fire — shaping whether STDP leads to LTP or LTD.
- **SOM+ cells** regulate apical input — which modulates whether top-down input reaches the spike threshold.
- **VIP+ cells** disinhibit SOM+, enabling **top-down driven STDP** during learning episodes.

So, interneurons also gate the conditions under which STDP occurs, based on behavioral state, attention, or surprise.



Key Features:

- **Bottom-Up Input (Layer 4)**:
  - Arrives **before** the pyramidal neuron fires.
  - **$\Delta t > 0$** (pre-before-post): Triggers **LTP (Long-Term Potentiation)** → synapse is **strengthened**.
  - This reinforces **causal input** from the environment.
- **Top-Down Input (Layer 1)**:
  - Arrives **after** the pyramidal neuron fired.
  - **$\Delta t < 0$** (post-before-pre): Triggers **LTD (Long-Term Depression)** → synapse is **weakened**.
  - ⊘ This reduces influence of **irrelevant or mistimed predictions**.
- **Neuron Firing**:
  - Occurs when bottom-up input leads to spiking.
  - Timing relationships with other inputs determine **whether synapses are strengthened or weakened**.

What This Diagram Shows:

- **STDP acts as a timing-sensitive learning rule**, deciding which inputs are **causally relevant**.
- It allows the cortex to **learn to trust sensory inputs** that predict spikes and **ignore or suppress unhelpful predictions**.

8. Sequences, composition, hierarchies

STDP and Temporal Sequences

Because STDP strengthens synapses when the presynaptic neuron fires before the postsynaptic neuron, it's naturally suited for learning temporal order — sequences of events.

How it works:

Imagine neurons A, B, and C fire in a consistent order (e.g. during a motor action or sensory experience):

- A → B → C
- STDP strengthens the synapse from A → B, then B → C, because in each pair, the presynaptic neuron fires just before the postsynaptic one.
- This forms a "chain" of activation: A can trigger B, and B can trigger C.

This is often called a synfire chain or sequence learning through temporal association.

Seen in:

- Hippocampus (e.g. during place cell sequences or replay)
- Motor cortex (movement planning)
- Auditory and visual cortex (processing of speech, gestures, etc.)

Learning Sequences = Building Blocks for Composition

Once neurons learn reliable temporal sequences, the brain can compose them into higher-level patterns — much like letters → words → phrases.

This leads to the idea of compositional learning.

Compositional Learning:

- Sequences become reusable building blocks.
- Neurons or assemblies that fire together get linked via Hebbian and STDP rules.
- Chunking: The brain can "package" frequently co-occurring events (e.g. a grip + wrist twist) into a composite pattern that can be reused as a single unit.

Hierarchical structure:

- Lower layers represent simple sequences (e.g., syllables).
- Higher layers represent combinations (e.g., words or actions).
- Even higher layers encode abstract rules or contexts (e.g., grammar, goals).

Current Theories and Models

Let's break down some of the leading hypotheses on how the brain learns sequences, composition, and hierarchies:

A. Recurrent Neural Circuits + STDP

- Cortical microcircuits with recurrent connections and STDP can self-organize into sequence-recognizing structures.
- Neurons learn to predict who comes next, and the timing of spikes reinforces temporal dependencies.
- Found in:
  - Hippocampus (CA3 recurrent network)
  - Prefrontal cortex (working memory traces)

B. Predictive Coding with Temporal Dynamics

- Predictive coding models can be extended to time: the brain doesn't just predict "what," but "what happens next."
- Prediction errors drive learning of sequences, using plasticity rules like STDP.
- Hierarchies allow different layers to operate on different time scales:
  - Low layers: fast features (syllables, small movements)
  - High layers: slow features (phrases, intentions)

This is called temporal predictive coding or hierarchical predictive coding.

C. Temporal Hebbian Models (e.g., SNNs, Liquid State Machines)

- Spiking Neural Networks (SNNs) use time-varying input + STDP to model sequence learning.
- In these models, neural dynamics act like memory traces, enabling recognition of time-dependent input patterns.
- These models can learn to compose patterns and generalize over time.

D. Chunking + Binding through Oscillations

- Brain rhythms (e.g., theta, gamma) gate plasticity and help group events into chunks.
- Theta phase precession in hippocampus may align sequences with learning windows, enabling STDP to link them.
- Oscillatory nesting (gamma in theta) may support multi-scale temporal binding — crucial for compositional structure.

Summary Table

Mechanism	Function in Learning
STDP	Encodes temporal order — basis of sequence learning
Recurrent connections	Sustain activity over time; enable chaining
Predictive coding	Learns “what happens next” — sequences + context
Hebbian chunking	Groups co-occurring sequences into composite units
Oscillatory dynamics	Time windows and binding — hierarchical composition

Final Insight

STDP is a low-level mechanism for temporal association, but it's the foundation upon which the brain builds:

- Sequences
- Predictive models of the world
- Hierarchies of concepts
- Flexible compositional thought

Temporal Predictive Coding and Hierarchical Predictive Coding

- how the brain predicts not just what, but what happens next across time and levels of abstraction.

It’s like learning to write by connecting letters into words — STDP writes the letters, and predictive, hierarchical models form the sentences.

What Is Predictive Coding, Recap

Before diving into the temporal and hierarchical aspects, let’s recall the basic principle:

Predictive coding is a theory where the brain continuously makes predictions about incoming sensory input and only processes the prediction errors — the differences between expected and actual input.

In cortex:

- Higher levels send predictions down.
- Lower levels compute errors (difference between input and prediction).
- Errors go up, updating the internal model.

What Is Temporal Predictive Coding?

Temporal Predictive Coding is the idea that:

The brain not only predicts what you perceive — it also predicts when it should happen.

Key Insight:

- The brain builds a temporal model of events.
- This allows it to:
    - Anticipate what will happen next.
    - Track temporal structure (speech, motion, melody).
    - Update predictions over time.

Mechanism:

- Prediction errors are not just spatial (e.g. a visual mismatch) — they're temporal mismatches too (e.g. something arriving earlier/later than expected).
- Neurons adapt not just their tuning, but also their timing expectations.
- STDP plays a key role, reinforcing correct timing relationships and weakening wrong ones.

Example:

- You expect the syllables in "hello" to arrive in a rhythmic pattern.
- If one is delayed, the temporal prediction error alerts your brain: something's wrong.
- You might interpret it as a different word — or infer emotion (e.g., hesitation).

What Is Hierarchical Predictive Coding?

Now add the vertical dimension:

In Hierarchical Predictive Coding, the brain uses multiple levels of abstraction, each predicting inputs at its own timescale and level of meaning.

Structure:

Hierarchy Level	Time Scale	Function
Low (e.g. V1)	Fast (ms)	Raw sensory input (edges, tones)
Mid (e.g. STG, MT)	Mid (100 ms)	Gestures, phonemes, movement chunks
High (e.g. PFC, TPJ)	Slow (seconds+)	Goals, context, syntax, task rules

Each level:

- Receives bottom-up error from the layer below.
- Sends top-down predictions to suppress expected input.
- Integrates over different time windows.

Example: Speech Perception

Imagine listening to someone say, "The cat sat on the mat."

Low levels:

- Predict timing of phonemes (e.g. /k/ in "cat").
- Process fast acoustic features and detect temporal deviations.

Mid levels:

- Predict word transitions (e.g., "the cat" → "sat").
- Learn common phrases through timing and co-occurrence.

High levels:

- Predict grammatical structure and meaning.
- Anticipate sentence completion based on context or intention.

If something breaks the expectation at any level, a prediction error is generated at that timescale and passed up.

Interaction Between Temporal and Hierarchical Levels

Now, combine both ideas:

- Temporal predictive coding operates within each layer.
- Hierarchical predictive coding coordinates layers over different timescales.

This results in a dynamic, multiscale system:

- Fast predictions (milliseconds) happen in sensory cortices.
- Slow predictions (seconds, context) come from higher cortex (e.g. prefrontal, parietal).
- All layers coordinate via prediction and error signals, shaping behavior and learning.

Models That Reflect This

Several computational and biological models are trying to capture this:

1. Deep Temporal Predictive Coding Networks

- Stack of predictive coding modules with learned time constants.
- Inspired by cortical hierarchy and used in modeling video prediction, motor control, language.

2. Active Inference (Friston)

- Adds action and planning to predictive coding.
- Hierarchies of models make predictions about future outcomes, not just passive perception.
- Temporal predictions are embedded in policies (sequences of actions).

3. Hawkins/HTM Models (Hierarchical Temporal Memory)

- Each cortical column models transition probabilities over time.
- Columns are arranged in hierarchies — compositional temporal learning.

• Strong emphasis on sequence memory + prediction.

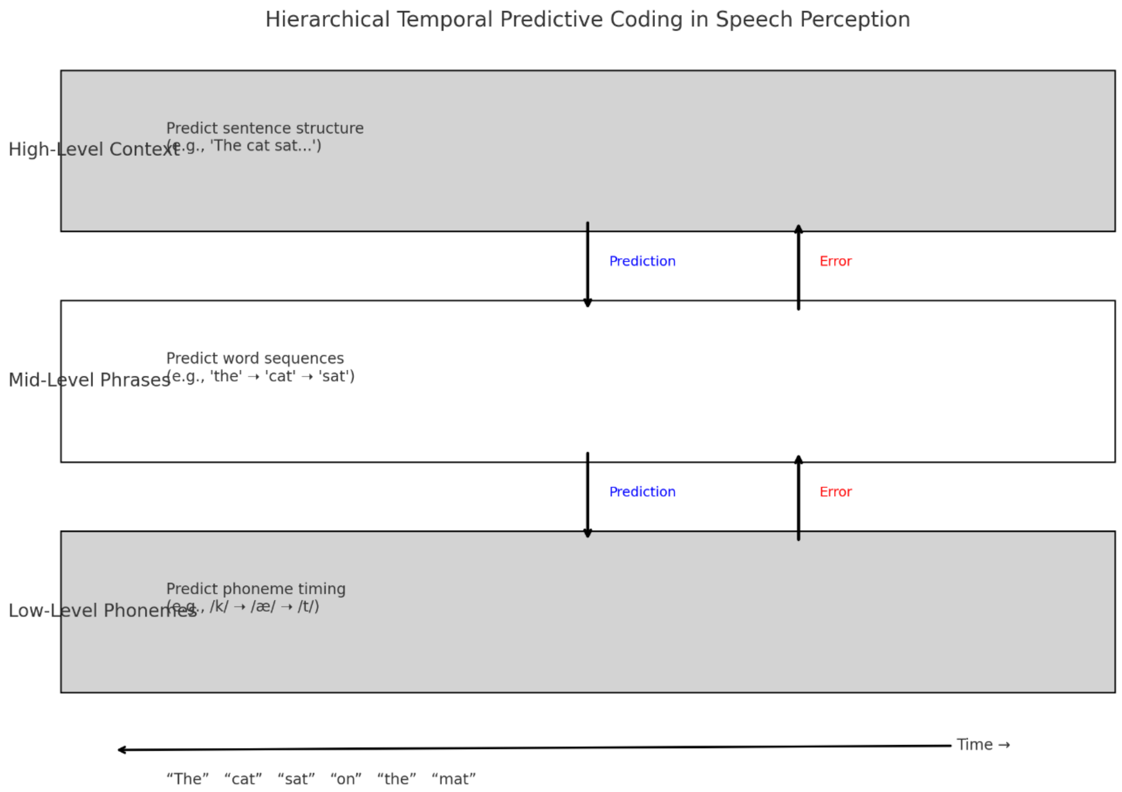
Summary Table

Feature	Temporal Predictive Coding	Hierarchical Predictive Coding
Focus	When inputs occur	What level of abstraction predicts what
Role	Learns sequences, rhythm, timing	Builds layered models across cortex
Key Mechanism	STDP, recurrent connections	Top-down + bottom-up loops
Output	Temporal prediction error	Multi-level error across time
Brain Basis	Cortical microcircuits + hierarchy	Visual, auditory, motor, prefrontal cortex

Final Insight

The brain is a multi-layer prediction engine, operating over time and abstraction. It uses STDP and local circuits to learn sequences, and hierarchical feedback to generalize and interpret.

Here's the diagram illustrating Hierarchical Temporal Predictive Coding in the context of speech perception:



Three Processing Layers:

1. High-Level Context (Top Layer)

    ◦ Predicts sentence structure and meaning (e.g., "The cat sat...")
    ◦ Operates on longer time scales (seconds)

2. Mid-Level Phrases
    ◦ Predicts word sequences (e.g., "the" → "cat" → "sat")
    ◦ Operates on mid-level time scales (hundreds of milliseconds)

3. Low-Level Phonemes
    ◦ Predicts individual sounds (/k/, /æ/, /t/)
    ◦ Operates on fast time scales (milliseconds)

Arrows:

- 🔵 Blue Arrows = Top-down predictions:

Each level predicts what the level below it should perceive next.

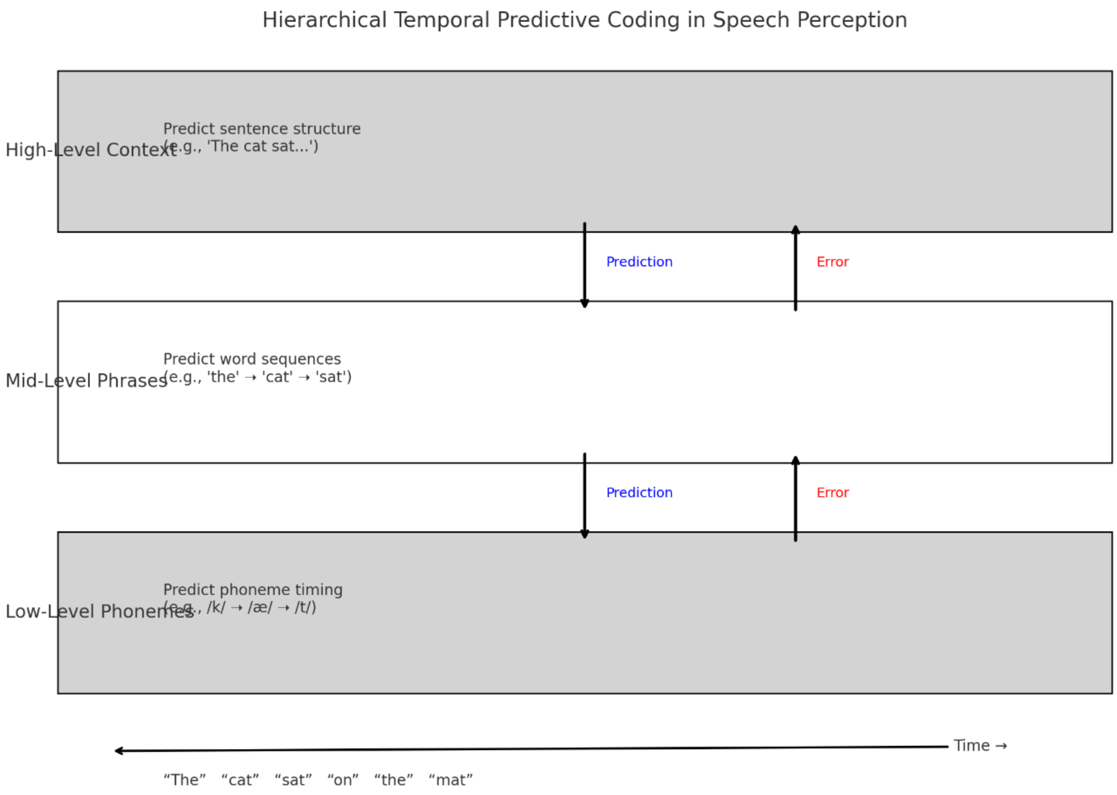
- 🔴 Red Arrows = Bottom-up error signals:

If reality doesn't match prediction, an error goes up to adjust the model.

Time Axis:

Speech unfolds in time — from phonemes to words to full sentences — and the brain predicts at multiple time resolutions simultaneously.

Here is the diagram mapping Hierarchical Temporal Predictive Coding to actual cortical brain regions involved in speech perception:



Mapped Cortical Areas:

1. Prefrontal Cortex
   - Deals with high-level goals, syntax, intentions.
   - Predicts sentence structure and context.
   - Operates on long time scales (seconds or more).
2. Temporal Lobe (Superior & Middle Temporal Gyri – STG/MTG)
   - Processes word sequences, semantic associations, grammar rules.
   - Intermediate timescales (hundreds of milliseconds).
3. Auditory Cortex (A1)
   - Encodes phonemes, pitch, and timing of sounds.
   - Operates on millisecond timescales.
   - The first stage of detailed auditory analysis.

Interactions:

- 🔵 Predictions flow top-down (blue arrows), shaping what lower areas expect to perceive.
- 🔴 Error signals flow bottom-up (red arrows), correcting mismatches and driving learning.

Real-Time Speech:

Each word in a sentence (e.g., "The cat sat on the mat") is processed across all layers:

- Low level: Matches phoneme timing and pronunciation.
- Mid level: Confirms word order and semantics.
- High level: Interprets the meaning, context, and predicts continuation.

9. Deep Temporal Predictive Coding Networks

Deep Temporal Predictive Coding Networks (DT-PCNs) are computational architectures that:

- Are structured in layers (deep),
- Operate over time (temporal),
- Learn by predicting their own future input and adjusting internal states based on prediction errors (predictive coding),
- Use recurrent or stateful dynamics to capture temporal context.

These models generalize predictive coding into a deep, temporal framework suitable for complex inputs like language, speech, and sequential reasoning.

Core Components

Component	Function
Hierarchical layers	Model increasing levels of abstraction (e.g., phoneme → word → phrase)
Recurrent dynamics	Allow each layer to remember and process sequences over time
Top-down predictions	Each layer predicts the activity of the layer below
Bottom-up errors	Discrepancies between predictions and actual inputs are passed upward
Temporal prediction	Predict the next time step, not just current input
Learning rule	Usually gradient descent on prediction error, possibly biologically inspired (e.g., Hebbian or STDP variants)

Language Processing in DT-PCNs

□ The challenge in language:

- Language is sequential: meaning depends on word order, timing, context.
- It is also hierarchical: letters → syllables → words → phrases → sentences → discourse.

DT-PCNs model this by having:

- Lower layers learn to predict short-term patterns (e.g., character or phoneme sequences).
- Middle layers capture medium-range syntax or grammar (e.g., word combinations, subject-verb agreement).
- Higher layers predict semantic content, context shifts, or discourse flow.

🔁 Example:

Suppose the model is trained on the sentence:

“The cat sat on the mat.”

Each layer learns to predict the next unit in time:

Layer	Predicts
Layer 1 (low)	Next character/phoneme (“t” → “h” → “e”)
Layer 2 (mid)	Next word based on short context (“the” → “cat”)
Layer 3 (high)	Sentence completion based on syntax/meaning (“sat on the mat”)

The prediction error at each time step adjusts weights at the layer that made the mistake, improving both temporal memory and compositional understanding.

Temporal vs Hierarchical Prediction in Language

Aspect	Temporal Prediction	Hierarchical Prediction
Purpose	Predict next item in time	Predict structure and meaning across layers
Challenge	Ambiguity, long dependencies	Compositionality, recursion
Solution in DT-PCN	Recurrent cells, error memory	Layered representation with feedback

DT-PCNs vs Transformers (e.g., GPT)

While transformers (like GPT) dominate language models today, DT-PCNs are more brain-inspired and may offer benefits like:

- Online learning (update model on-the-fly using prediction errors),
- Energy-efficient inference (only compute differences),
- Local learning rules (closer to biological plausibility),
- Interpretability via clear hierarchical prediction chains.

They’re still in development but are highly promising in neuroscience-inspired AI, especially in areas needing temporal understanding and incremental prediction.

Real-World Applications

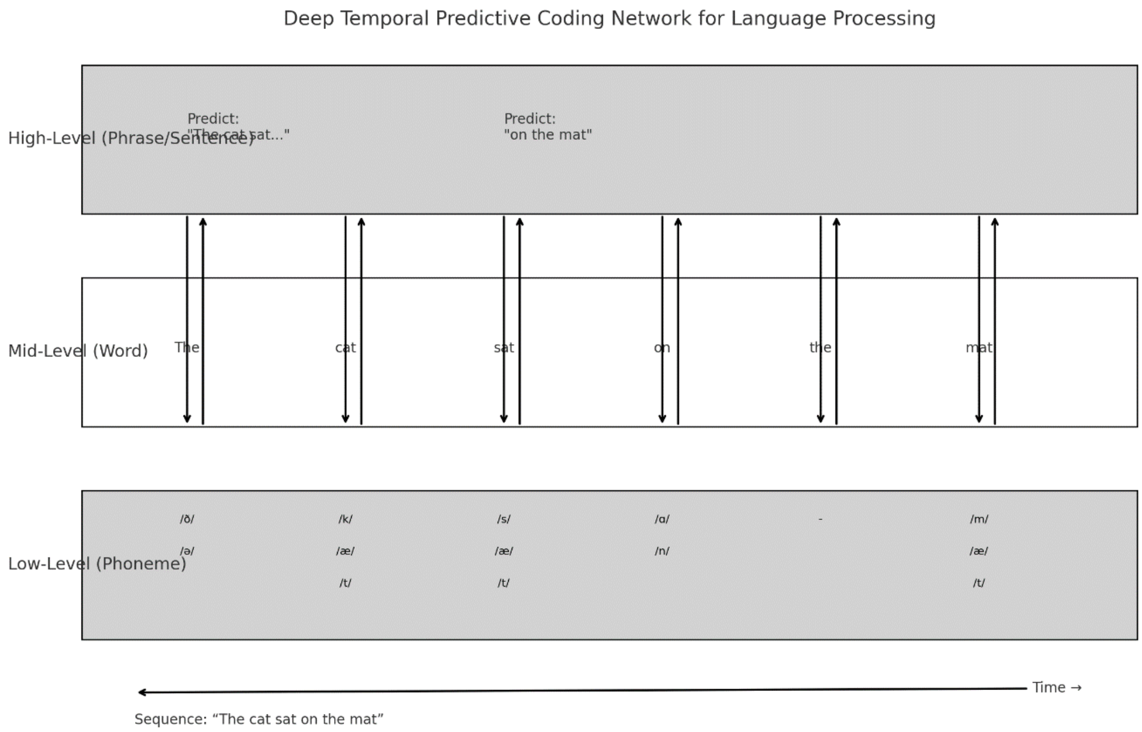
DT-PCNs (or similar temporal predictive coding models) have been applied to:

- Speech recognition and generation (e.g., learning phoneme transitions, prosody),
- Text prediction and correction (e.g., real-time language modeling),
- Language grounding (predicting language based on visual or motor inputs),
- Sequential decision making (e.g., conversational agents predicting user intent over time).

Summary

Feature	Deep Temporal Predictive Coding Network
Inspiration	Brain’s predictive coding, layered cortical hierarchy
Temporal modeling	Predicts next input in time; remembers prior state
Hierarchy	Each layer predicts the one below at its own abstraction level
Language role	Models phoneme-to-word-to-sentence dynamics via prediction
Learning	Minimizes prediction error; local rules possible (e.g., Hebbian/STDP variants)
Brain analogs	Auditory cortex, STG, Broca’s area, prefrontal cortex, etc.

Here’s the diagram showing how a **Deep Temporal Predictive Coding Network (DT-PCN)** processes a sentence across time and layers:



Hierarchical Layers:

1. High-Level (Phrase/Sentence)
   - Predicts sentence-level meaning:

→ e.g., "The cat sat..." → "on the mat"

   - Operates over longer temporal spans

2. Mid-Level (Word)
   - Predicts the next word in the sequence

→ e.g., "the" → "cat" → "sat"...

3. Low-Level (Phoneme)
    - Predicts phoneme sequences

→ e.g., /k/ → /æ/ → /t/ for "cat"

Arrows:

- 🔵 Top-down blue arrows:

Show how each layer predicts the layer below.

- 🔴 Bottom-up red arrows:

Carry prediction errors upward when mismatches occur.

Time Axis:

As the sentence unfolds over time (left to right), each layer:

- Makes predictions based on past input,
- Updates internal representations,
- Refines future predictions.

This reflects how a DT-PCN learns and processes language incrementally, combining sequence memory and hierarchical abstraction — just like your brain does when you listen or read.

What Is Online Learning in DT-PCNs?

Online learning means the model updates its internal state and weights continually as data arrives — in real time, rather than all at once (as in batch training).

In a DT-PCN:

- Each layer predicts its own input at each time step.
- The prediction error is computed immediately.
- A local update rule (like gradient descent or Hebbian-like learning) adjusts weights and activations on the fly.

This is closer to how the brain learns:

- You don't need to store all past data.
- You don't need to know when the sequence ends.
- Learning is continuous, real-time, and adaptive.

What Changes During Online Learning?

Component	What Changes
Synaptic weights	Adjusted using local prediction errors (e.g. $\Delta w \propto$ error $\times$ presynaptic activity)
Hidden state	Updated recurrently — carries forward memory of recent past
Predictions	Improve over time, reduce error if structure is learned
Confidence	Can be encoded via precision weighting (low error → high certainty)

Example: Language Sentence Stream

As a DT-PCN processes a sentence like:

"The cat sat on the mat"

It may initially mispredict "on" after "sat", but after encountering similar phrases, it adjusts:

- Next time it hears "The cat sat...", it will predict "on" more confidently.
- This reflects sequence learning, error-driven adaptation, and temporal context encoding.

Attention in DT-PCNs

Attention modulates which inputs or internal representations the model focuses on.

In DT-PCNs, attention is implemented by modulating:

- The gain (amplification) of certain inputs or prediction errors,
- The precision weighting — how much error from each layer or input channel affects the update.

Mechanisms of Attention

✅ A. Precision-weighted prediction errors

- Inspired by Friston's predictive coding models.
- Errors are scaled by a learned or context-driven precision term.
- More reliable signals get prioritized (e.g., a clear syllable in noise).

✅ B. Top-down attention signals

- High-level layers (e.g., prefrontal) send attention control signals.
- These bias lower-layer predictions toward task-relevant patterns (e.g., listening for a name in a crowd).
- This is similar to modulatory input in biological cortex (e.g., cholinergic or dopaminergic signals).

✅ C. Dynamic gating of recurrent state

• Some attention mechanisms modulate the recurrent memory gate, deciding how much of the past is carried forward.

Combined: Online Learning + Attention

Feature	Effect in DT-PCN
Online learning	Allows real-time adaptation to input structure
Attention mechanisms	Focus learning and prediction on most relevant signals
Precision weighting	Controls which errors drive learning
Top-down bias	High-level context guides low-level interpretation (e.g., disambiguating homophones)

Example Use Case: Speech in Noisy Environment

- DT-PCN listens to: "The... [noise] ...mat"
- Initially mispredicts the missing word.
- If attention is focused (e.g., user is actively listening), the model:
  - Upweights reliable error signals,
  - Downweights ambiguous noise,
  - Adapts more strongly to context ("sat on the...").

Over time, it learns to disambiguate based on sentence-level expectations.

Visual Summary (Conceptual)

[High-Level Context]

⇅ attention bias

[Mid-Level Sequences]

⇅ prediction & error

[Low-Level Signals]

⇅ prediction & error

⇄ Online updates (weights + state)

- Prediction flows top-down.
- Errors flow bottom-up.
- Attention modulates prediction strength and learning rate.

Scenario: Real-Time Speech Input

The model is listening to:

"The cat sat on the mat."

Let’s say this sentence is coming in word by word, one at a time, just like it would in a conversation.

Baseline: No Attention Modulation

Without attention, the DT-PCN treats each input equally — it:

- Makes predictions based on prior learned statistics.
- Updates weights based on prediction errors uniformly.

🎁 Internal Processing:

Time Step	Input Word	Prediction	Error	Learning
t1	"The"	—	—	initializes context
t2	"cat"	"dog"	●	small update
t3	"sat"	"ran"	●	small update
t4	"on"	"under"	●	small update
t5	"the"	"a"	●	small update
t6	"mat"	"chair"	●	small update

The model makes mistakes at each step (it's still learning), but the updates are modest, because it treats all errors as equally informative.

Now Add Attention

Let’s suppose the user is now actively focused on understanding the location of the cat. This attention is modeled as a top-down signal:

- Boosts precision weighting on location-related words (e.g., "on", "mat").
- Increases learning rate for contextually relevant predictions.

⚙️ How This Changes Processing:

Time Step	Input Word	Prediction	Error	Learning	Attention Effect
t1	"The"	—	—	context initialized	—
t2	"cat"	"dog"	●	small update	neutral
t3	"sat"	"ran"	●	small update	neutral
t4	"on"	"under"	●	▲ larger update	attention amplifies error

t5	"the"	"a"	●	moderate update	neutral
t6	"mat"	"chair"	●	▲ larger update	attention amplifies update

Resulting Behavior

With attention:

- The model learns faster from attended information.
- Errors in attended words cause larger weight updates.
- Predictions in similar future sentences become more accurate, especially around location terms.

For example, next time it hears:

"The dog sat on the..."

It might correctly predict "mat" — because the attention-modulated learning helped strengthen that prediction path.

Biological Analogy

This reflects real brain behavior:

- When you pay attention, neuromodulators (like acetylcholine or norepinephrine) increase the gain on relevant cortical circuits.
- This results in more plasticity (learning) in attended pathways.

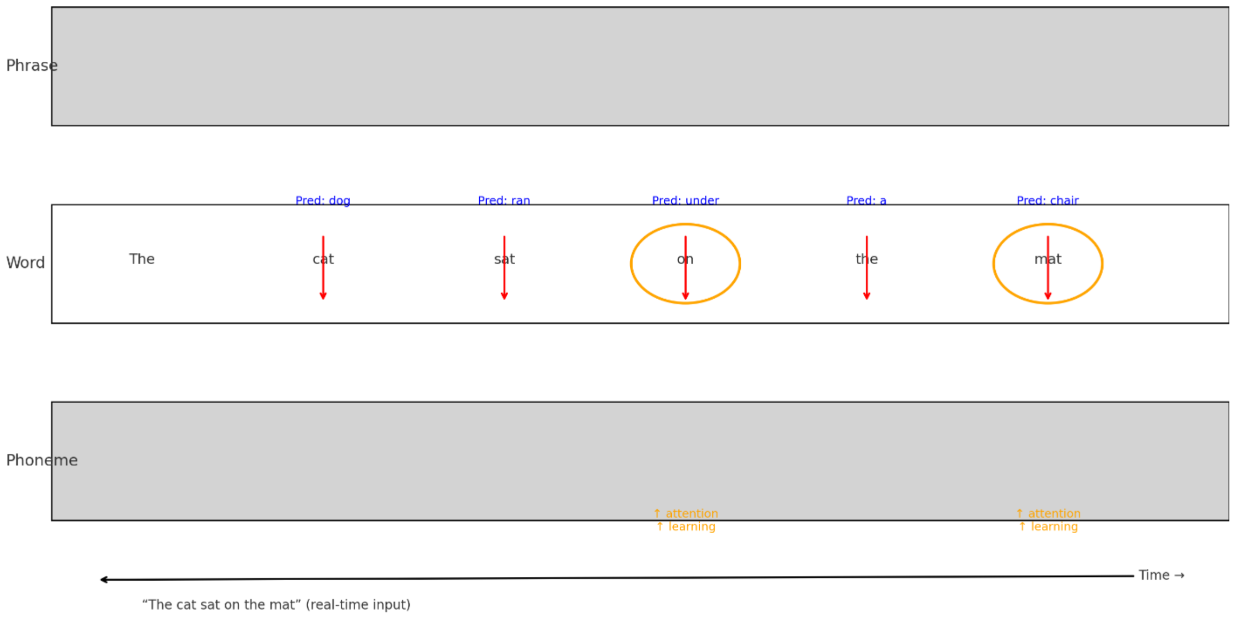
Key Differences With and Without Attention

Feature	No Attention	With Attention
Error weighting	Uniform	Precision-scaled
Learning	Equal across tokens	Focused on relevant words
Prediction improvement	Slow	Faster in attended domains
Generalization	Broad, shallow	Deep in relevant context

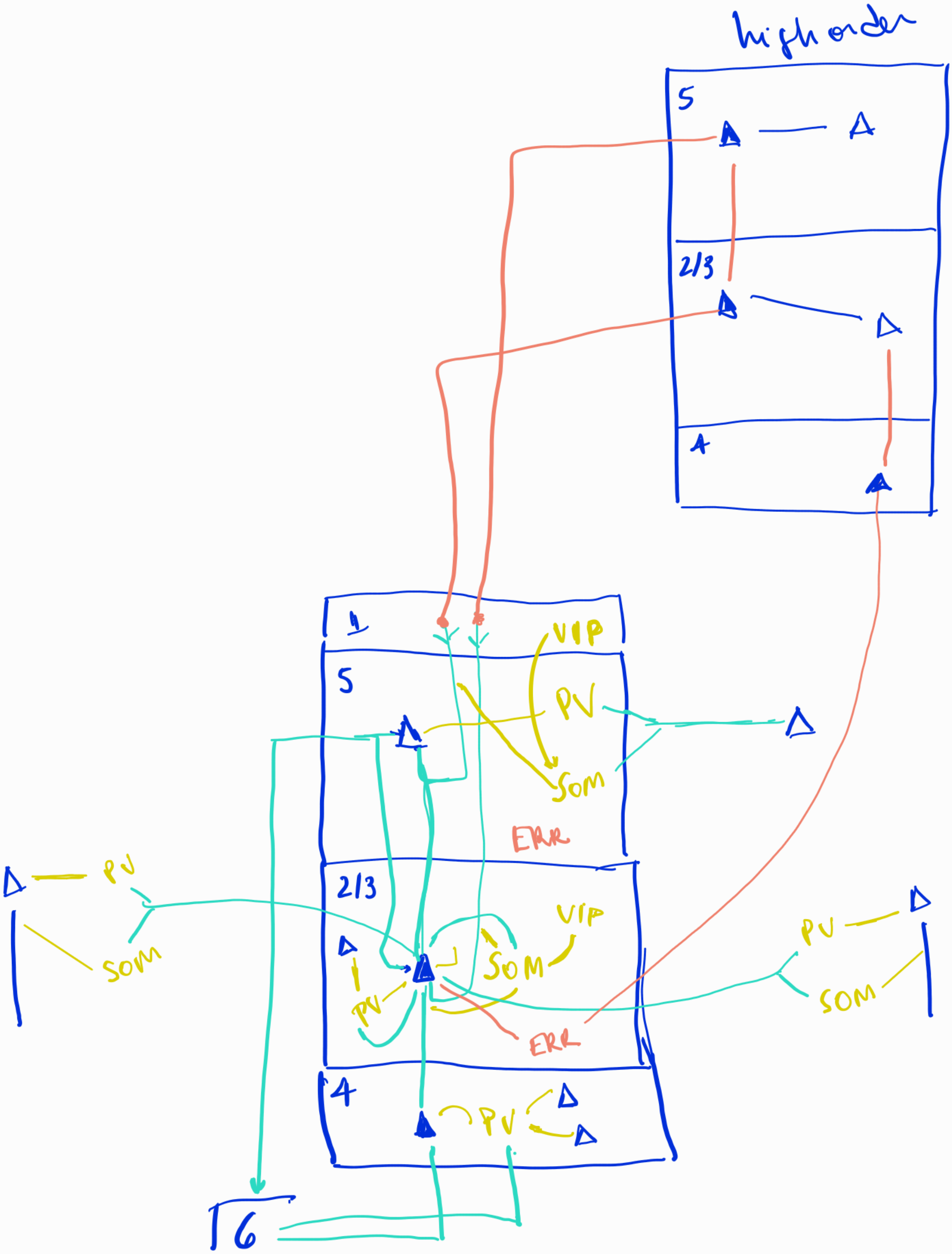
Summary

A DT-PCN, when combined with attention, doesn't just passively process input — it prioritizes and learns strategically, like a human listener who focuses on meaning and context.

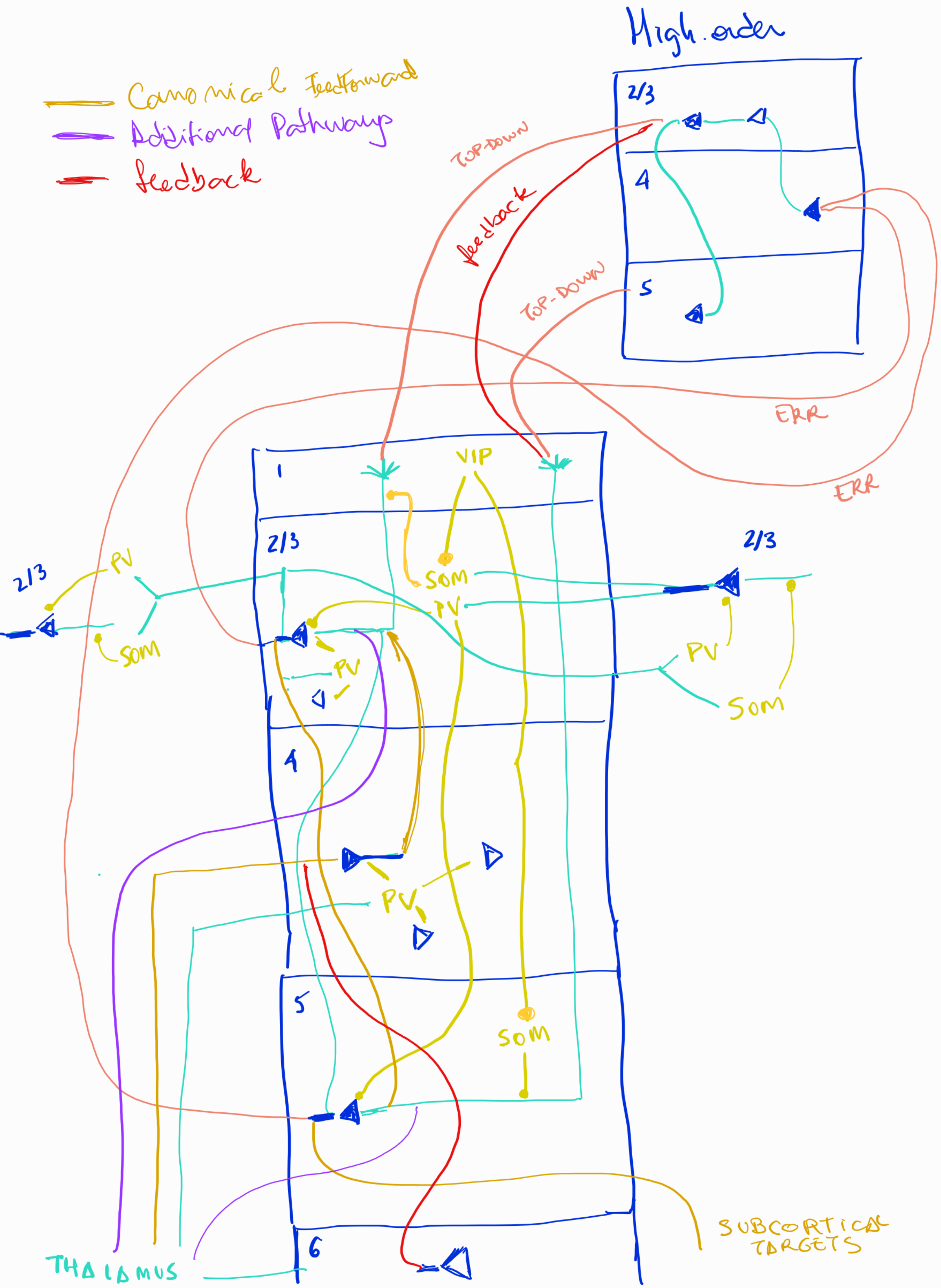
DT-PCN Real-Time Processing with Prediction, Error, and Attention



Modelo



high order
5
2/3
4
1
5
VIP
PV
SOM
ERR
2/3
VIP
SOM
PV
ERR
4
PV
PV
SOM
PV
SOM
6



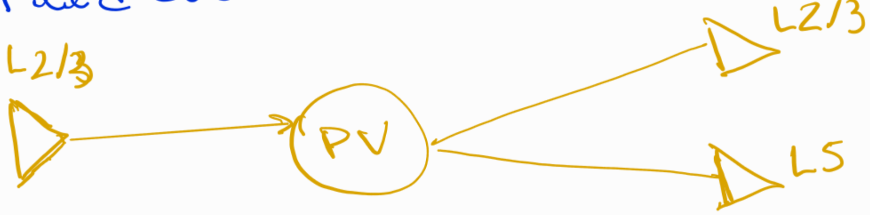
Canonical Feedforward
Additional Pathways
feedback
High order
2/3
4
5
TOP-DOWN
feedback
TOP-DOWN
ERR
ERR
1
VIP
2/3
SOM
PV
PV
2/3
PV
SOM
PV
SOM
2/3
PV
4
PV
5
SOM
6
THALAMUS
SUBCORTICAL TARGETS

24/09/2025 - Interneuron Circuit Motifs

PV Circuit Functions:

- FeedForward Inhibition:

- FeedBack Inhibition



- Lateral inhibition: cross-column
- Temporal control
  - precise spiking time
  - geração de ritmo gamma

SOM Circuit Functions

- Top-down gating
  - controla apical dendritic integration
- Attention
  - suprime sinais top-down irrelevantes
- Cross-layer inhibition

- Long-range inhibition
  - Horizontal connections between columns

VIP Circuit Functions

- Attention enhancement

  VIP → SOM → disinhibitory pathway

- Learning facilitation
  - Context-dependent plasticity gating
- State-dependent modulation
  - Different effects across brain states
- Rapid disinhibition
  - Fast synaptic dynamics